

TANKHOY, Russian Federation

WMO: 308240

Lat: 51.555N

Lon: 105.113E

Elev: 472

StdP: 95.78

Time Zone: 8.00 (E08)

Period: 94-06

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-29.4	-26.2	-31.1	0.2	-29.2	-28.0	0.3	-25.8	6.3	-11.9	5.1	-9.5	1.6	180	0.656

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.7	22.6	17.3	21.0	16.6	19.7	16.0	18.5	21.1	17.5	20.0	16.6	19.0	2.1	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.5	13.2	20.3	16.4	12.3	19.0	15.5	11.6	18.1	54.7	20.9	51.2	20.2	48.4	19.1	22.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.3	6.1	5.1	DB	-31.3	26.4	4.5	2.3	-34.5	28.1	-37.2	29.4	-39.7	30.7	-42.9	32.4	(4)
(5)			WB	-31.2	20.1	4.5	1.5	-34.5	21.1	-37.1	22.0	-39.7	22.8	-43.0	23.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.6	-15.1	-16.2	-9.0	0.5	6.7	11.6	15.8	14.6	8.8	2.4	-4.3	-9.4	(6)	
	DBStd	11.50	5.14	5.74	5.78	3.69	3.80	3.06	2.60	2.19	2.90	3.70	4.42	3.42	(7)	
	HDD10.0	3842	779	733	590	284	117	17	0	1	56	238	428	601	(8)	
	HDD18.3	6470	1038	966	848	534	361	201	87	118	285	496	678	859	(9)	
	CDD10.0	423	0	0	0	0	15	66	178	143	21	1	0	0	(10)	
	CDD18.3	9	0	0	0	0	0	1	7	2	0	0	0	0	(11)	
	CDH23.3	30	0	0	0	0	2	4	23	1	0	0	0	0	(12)	
	CDH26.7	3	0	0	0	0	0	0	3	0	0	0	0	0	(13)	
(14) Wind	WSAvg	2.2	2.1	1.7	2.0	2.2	2.3	2.1	1.9	2.0	2.4	2.5	2.7	2.6	(14)	
(15) Precipitation	PrecAvg	781	24	11	18	47	53	99	148	161	101	40	41	37	(15)	
	PrecMax	977	50	17	37	100	91	238	271	247	215	77	92	58	(16)	
	PrecMin	624	9	3	8	14	24	14	57	98	29	8	20	11	(17)	
	PrecStd	115	10	4	8	26	21	51	64	48	50	16	17	15	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.0	0.1	7.7	14.0	22.1	23.4	25.9	22.5	17.2	15.1	7.1	2.9	(19)	
		MCWB	-1.7	-3.0	2.4	6.6	12.4	14.9	18.9	17.5	12.8	8.3	3.2	-0.3	(20)	
	2%	DB	-2.5	-2.7	3.9	10.2	18.3	20.6	23.4	21.1	16.0	11.0	5.0	-0.2	(21)	
		MCWB	-3.9	-4.8	0.2	4.8	10.8	14.6	18.6	17.1	13.1	7.3	2.2	-2.0	(22)	
	5%	DB	-5.5	-5.1	1.8	7.5	15.7	18.4	21.9	20.0	15.1	9.1	3.5	-2.2	(23)	
		MCWB	-6.7	-6.4	-0.9	3.3	9.3	13.7	17.6	16.8	12.3	6.3	1.2	-3.7	(24)	
	10%	DB	-7.7	-7.1	0.0	5.5	13.5	16.8	20.3	18.9	13.9	7.7	1.8	-4.0	(25)	
		MCWB	-8.6	-8.2	-2.1	2.0	8.2	12.8	16.8	16.0	11.3	5.4	0.1	-5.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.3	-2.2	2.5	7.6	13.6	16.7	20.5	19.1	15.1	8.9	3.8	0.4	(27)	
		MCDB	-0.3	-0.1	7.0	12.0	20.7	20.5	23.7	20.9	16.0	12.7	6.1	1.5	(28)	
	2%	WB	-3.6	-4.7	0.7	5.1	11.7	15.6	19.3	17.9	13.7	7.8	2.5	-1.9	(29)	
		MCDB	-2.4	-2.9	3.4	10.0	16.5	18.9	22.4	20.2	15.4	10.5	4.6	-0.2	(30)	
	5%	WB	-6.6	-6.4	-0.8	3.7	10.3	14.6	18.1	17.1	12.6	6.6	1.4	-3.7	(31)	
		MCDB	-5.6	-5.1	1.5	7.1	14.6	17.5	20.9	19.3	14.6	8.9	3.1	-2.4	(32)	
	10%	WB	-8.6	-8.1	-1.9	2.4	8.7	13.6	17.2	16.3	11.6	5.4	0.2	-5.1	(33)	
		MCDB	-7.7	-7.1	0.0	5.1	12.8	16.2	19.8	18.5	13.6	7.5	1.9	-4.1	(34)	
(35) Mean Daily Temperature Range	MDBR		8.7	10.5	10.7	7.8	8.9	7.8	6.7	6.3	6.9	7.0	7.2	7.7	(35)	
	5% DB	MCDBR	10.0	12.3	12.3	11.4	12.8	10.8	9.5	8.0	8.3	9.2	8.9	9.4	(36)	
		MCWBR	8.4	10.3	8.9	7.0	6.9	6.0	5.2	4.9	5.6	6.5	7.0	7.6	(37)	
	5% WB	MCDBR	9.3	12.1	11.2	10.6	11.7	8.9	8.3	7.0	7.5	8.8	8.5	8.9	(38)	
		MCWBR	8.5	10.3	8.5	6.8	7.1	5.6	5.2	4.9	5.7	6.7	6.9	7.4	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.274	0.301	0.342	0.391	0.456	0.451	0.440	0.402	0.346	0.322	0.292	0.275	(40)	
	taud		2.100	2.075	2.099	2.167	2.036	2.176	2.259	2.347	2.454	2.400	2.179	2.070	(41)	
	Ebn at Noon		714	801	840	843	799	808	811	826	842	788	687	633	(42)	
	Edh at Noon		76	105	126	133	160	141	128	111	89	77	70	67	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.28	2.33	3.63	4.60	5.56	5.80	5.37	4.52	3.55	2.22	1.27	0.91	(44)	
	RadStd		0.06	0.12	0.20	0.32	0.30	0.34	0.37	0.29	0.35	0.17	0.08	0.07	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (10 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page